

BO I VESTERALEN, Norway

WMO: 011560

Lat: 68.607N

Lon: 14.433E

Elev: 25

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.3	22.4	9.4	8.9	22.3	11.9	10.1	23.9	45.2	37.7	40.3	35.3	10.0	70	0.787

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	69.0	59.0	65.6	57.3	62.6	55.8	60.2	66.7	58.2	64.1	56.6	61.5	8.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.2	69.9	62.7	55.4	65.4	60.2	53.9	62.0	58.5	26.5	66.7	25.2	64.3	24.2	61.6	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.1	31.5	27.5	17.2	75.0	2.7	4.2	15.3	78.0	13.7	80.5	12.3	82.9	10.3	85.9
			15.1	63.4	2.4	2.9	13.4	65.5	12.0	67.2	10.7	68.8	8.9	70.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	32.9	32.1	32.9	37.7	44.4	48.9	54.8	54.3	50.1	43.3	38.7	35.8
	DBStd	9.43	5.39	5.30	5.10	4.65	5.26	5.01	5.12	4.50	4.53	5.15	5.08	4.99
	HDD50	3255	530	502	530	369	188	78	6	9	53	212	339	439
	HDD65	8319	995	922	995	818	638	484	322	333	448	671	789	904
	CDD50	419	0	0	0	1	15	45	155	142	56	5	0	0
	CDD65	8	0	0	0	0	0	1	6	1	0	0	0	0
	CDH74	15	0	0	0	0	0	2	12	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.5	14.5	14.5	14.7	12.1	11.1	10.4	8.9	9.1	11.2	13.4	14.7	15.7
Precipitation	PrecAvg	43.5	3.8	3.7	3.5	2.7	2.5	2.1	2.4	3.0	4.3	5.7	4.8	5.3
	PrecMax	54.0	8.9	8.2	6.7	4.7	4.3	5.0	6.7	8.0	9.1	10.3	12.2	9.4
	PrecMin	33.0	0.0	1.9	0.6	0.6	0.7	0.0	0.3	1.1	1.5	1.1	0.0	0.4
	PrecStd	6.3	2.1	1.4	1.7	1.0	0.9	1.3	1.6	1.5	1.9	2.4	2.8	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	43.3	44.8	56.5	65.4	70.8	75.4	70.4	63.7	55.8	50.6	46.5
		MCWB	41.9	41.0	40.2	47.0	55.2	57.7	64.6	60.3	55.3	49.7	48.2	44.2
	2%	DB	43.2	42.1	42.7	50.0	58.9	64.9	71.1	66.8	60.8	53.7	48.6	45.4
		MCWB	40.7	40.2	40.2	43.2	50.3	55.3	60.7	58.5	53.8	48.7	46.0	43.2
	5%	DB	42.0	41.0	41.4	45.5	55.3	60.2	67.1	64.3	58.5	51.6	47.3	44.3
		MCWB	39.5	39.0	39.0	40.9	48.5	52.5	58.3	57.3	52.5	48.5	44.5	41.9
	10%	DB	40.5	39.8	40.3	43.5	51.9	56.6	63.1	61.5	56.2	50.1	45.9	42.9
		MCWB	37.9	37.4	37.8	39.9	45.4	50.9	56.6	55.9	51.8	47.3	43.1	40.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	42.3	41.8	47.6	55.5	59.0	66.5	62.0	56.7	51.6	49.1	45.5
		MCDB	43.9	42.8	42.8	56.4	65.0	68.4	74.4	67.1	61.8	53.7	50.3	46.0
	2%	WB	41.4	40.6	40.5	43.7	51.5	56.1	61.9	59.6	54.7	50.3	47.0	44.0
		MCDB	43.0	41.8	42.1	48.0	58.0	63.6	69.1	65.2	58.7	52.4	48.4	45.1
	5%	WB	40.0	39.5	39.6	42.1	48.5	53.7	58.8	58.1	53.7	49.0	45.1	42.2
		MCDB	41.8	41.0	41.5	45.0	54.2	59.1	65.4	63.2	57.1	51.1	46.9	44.1
	10%	WB	38.1	37.8	38.0	40.8	46.7	51.6	57.0	56.5	52.6	47.7	43.4	40.4
		MCDB	40.3	39.6	40.2	43.2	50.9	56.1	62.2	60.7	55.7	50.0	45.8	42.7
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.8	6.0	7.0	8.0	7.7	8.6	7.9	6.8	5.6	5.7	5.7
		MCDBR	5.9	6.2	5.6	9.4	13.3	13.9	14.9	12.5	9.4	5.6	6.1	6.6
		MCWBR	5.7	6.0	5.1	6.0	7.4	8.0	7.9	6.7	5.6	4.5	6.2	6.6
	5% WB	MCDBR	6.0	6.6	5.7	7.9	12.3	12.4	13.8	10.8	7.2	5.1	6.0	6.8
		MCWBR	6.2	6.6	5.4	5.9	7.1	7.7	8.1	6.3	5.3	4.9	6.6	7.1
Clear-Sky Solar Irradiance	taub	0.371	0.293	0.327	0.342	0.346	0.344	0.348	0.341	0.327	0.295	0.352	N/A	
	taud	2.511	2.140	2.256	2.380	2.453	2.502	2.518	2.526	2.513	2.244	2.499	N/A	
	Ebn at Noon	23	148	219	255	268	272	267	256	227	147	20	0	
	Edh at Noon	3	14	23	28	29	28	27	24	19	12	3	0	
All-Sky Solar Radiation	RadAvg	7	96	438	978	1411	1551	1410	1021	539	179	21	0	
	RadStd	1	22	62	86	108	223	147	96	50	33	2	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page